

HAT ISLAND, NU, Canada

WMO: 710840

Lat: 68.320N

Lon: 100.090W

Elev: 36

StdP: 100.89

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-40.4	-39.2	N/A	N/A	N/A	N/A	N/A	N/A	19.0	-28.3	16.8	-26.8	5.8	270	1.230

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.0	14.6	N/A	12.1	N/A	10.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	5.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.9	15.0	13.3	DB	-41.9	18.7	2.1	2.5	-43.5	20.5	-44.7	22.0	-45.9	23.4	-47.5	25.3	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-13.0	-30.6	-31.5	-27.9	-19.9	-8.6	1.2	5.9	5.5	1.0	-6.7	-18.4	-26.7	(6)
(7)		DBStd	14.67	5.37	5.38	6.29	5.81	5.22	3.46	3.13	3.14	3.28	4.61	5.96	6.10	(7)
(8)		HDD10.0	8393	1259	1163	1173	896	578	265	136	144	270	518	853	1136	(8)
(9)		HDD18.3	11421	1518	1397	1432	1146	836	514	387	397	520	777	1103	1394	(9)
(10)		CDD10.0	14	0	0	0	0	0	1	7	6	0	0	0	0	(10)
(11)		CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	7.1	7.3	7.1	7.1	7.1	6.9	6.4	6.2	6.5	7.7	8.2	7.8	7.4	(14)
(15)	Precipitation	PrecAvg	169	6	5	8	7	8	14	26	29	27	19	9	9	(15)
(16)		PrecMax	236	21	13	40	17	24	37	53	56	57	44	25	21	(16)
(17)		PrecMin	107	1	1	2	2	1	3	11	13	14	5	2	1	(17)
(18)		PrecStd	36	5	3	8	3	5	8	10	9	11	9	5	5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-13.4	-13.9	-9.1	-2.2	2.2	14.8	19.0	17.5	10.6	2.1	-2.3	-6.3	(19)
(20)		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(20)
(21)		2%	DB	-16.2	-16.9	-12.5	-5.7	0.7	10.7	15.4	14.1	8.2	0.8	-5.6	-11.2	(21)
(22)		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(22)
(23)		5%	DB	-19.7	-19.4	-14.8	-8.2	0.0	7.8	12.7	11.8	6.7	0.0	-7.6	-14.3	(23)
(24)		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(24)
(25)		10%	DB	-22.7	-22.6	-17.8	-10.9	-1.3	5.6	10.7	9.8	5.5	-0.9	-9.9	-17.5	(25)
(26)		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(27)
(28)		MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(28)
(29)		2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(29)
(30)		MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(30)
(31)		5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(31)
(32)		MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(32)
(33)		10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(33)
(34)		MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(34)
(35)	Mean Daily Temperature Range	MDBR	5.5	5.7	6.5	7.4	5.7	4.2	6.0	4.4	2.9	3.4	5.3	5.5	(35)	
(36)		5% DB	MCDBR	8.6	9.1	8.6	8.5	5.4	8.1	9.7	7.7	4.7	3.1	6.5	8.8	(36)
(37)		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(37)
(38)		5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(38)
(39)		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.150	0.207	0.238	0.292	0.313	0.327	0.357	0.329	0.301	0.245	0.156	N/A	(40)	
(41)		taud	1.873	2.042	2.175	2.051	2.045	2.166	2.263	2.509	2.514	2.207	1.891	N/A	(41)	
(42)		Ebn at Noon	126	605	814	845	868	866	823	820	753	543	97	0	(42)	
(43)		Edh at Noon	7	46	76	118	134	123	109	77	59	39	6	0	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.03	0.49	2.09	4.47	6.53	6.71	5.66	3.58	1.69	0.54	0.09	0.00	(44)	
(45)		RadStd	0.00	0.03	0.09	0.18	0.32	0.39	0.36	0.27	0.21	0.09	0.01	0.00	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
(46)	Station Only	(d) N/A	(e) +1.28	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page